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Problem Chosen

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2003 Mathematical Contest in Modeling (MCM) Summary Sheet

(Attach a copy of this page to each copy of your solution paper.)

Type a summary of your results on this page. Do not include the name of your school, advisor, or team members on this page.

Summary

New laws will soon mandate 100% screening of all checked bags at the 429 airports throughout the United States by explosive detection systems (EDSs). Under this condition, we are tasked to determine the number of EDSs needed and the strategy to schedule the flights. Moreover, the necessity of the supplements with explosive trace detection (ETD) machines should be studied.

We consider tasks 1 and 3 as a major problem and a model of queue service system could represent it. By studying a general queue service system, we obtain the relationship between number of passengers and devices required. Furthermore, criteria for evaluating the queue service system are developed. Security of flight, expenditure of airport and average time for passengers waiting are primary factors that determine the quality of service.

Adapting the conclusions obtained to Airports A & B, we develop a feasible strategy to schedule the flights. Using both theoretical method and dynamical computer simulation, we obtain that 41 EDSs are required in Airport A and the average time spent on waiting is 14.865 minutes per passenger; 44 EDSs are required in Airport B and the average time spent on waiting is 14.473 minutes per passenger.

Based on the analysis of the conclusions and results obtained, to accomplish task 2, 4, and 6, we write a position paper on security-related objectives, a recommendation about the screening process and a memo explaining how to adapt our models to other airports and flights.

To fulfil task 7, performances of both EDSs and ETDs are concentrated. We find that ETDs have higher accuracy while costing more money. It is not economical to use only ETDs for screening, but they could be a helpful supplement for EDSs.

Finally, we discuss the way to best fund future scientific research.

To Screen or Not?

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Problem restatement and analysis

Transportation Security Administration (TSA) will soon begin screening 100% of checked baggage at all 429 commercial airports across the United States. It uses two kinds of screening systems: explosive detection system (EDS) and explosive trace detection (ETD). EDS create three-dimensional images of a bag by x-rays, showing the density of each item. But it cost nearly \$1 million to purchase and thousands of dollars to install. Because of the high cost and limited number available, we need to do a detailed analysis to best utilize EDS. ETD is another kind of screening systems with higher accuracy but higher labor cost. Each system has its advantages and disadvantages. Our objective is to develop a method to best use of them synthetically.

Basic Assumptions

1. Airport fact book [1] shows that busy airports such as Los Angeles Airport and Chicago O'Hare Airport can reach an annual aircraft movements of 800,000. Because Airports A & B is the largest ones in the Midwest Region, it is reasonable to assume that the data given in Appendix A is during exactly one hour of the busy time. We assume that it is between 2pm and 3pm.
2. Flights depart every 5 minutes.
3. The number of flights departing within 5 minutes is not greater than 4.
4. Check-in process begins 90 minutes and ends 30 minutes before departure.
5. Only when all passengers have checked in can a flight depart and if the examinations could not be finished 30 minutes prior the departure time, the flight will delay.

Variables and Notations

p Device accuracy

p_e Device error probability

M_i $i = 1, 2, \dots, 8$ Types of flights

L_i Length of queue i

w Time on waiting

I_{ij} The interference degree from queue waiting for flight i to queue waiting for flight j .

$S_{i,j}$ Flight of type i departing at time j

r_i Seat occupation of flight i

C_i Number of seats on flight i

v Speed of screening

Model I: Screening with EDSs

Mathematic Model of One-Counter Service

As a basic model, we consider that there is only one check counter and all the passengers discussed in this model would depart at the same time. To investigate a general regulation, suppose that passengers arrive at the airport between times t_1 and t_2 . Considering the bags come at the same time, we assume that arrival time of bags T is a random variable whose probability density function is $f(t)$ and probability distribution function is $F(t)$. Let Q be the total number of bags arriving, and Q_a be the number of bags arriving we have:

$$\begin{aligned} Q_a(t) &= \int_{-\infty}^t Qf(t)dt \\ &= QF(t) \end{aligned} \quad (1)$$

The bags arriving trend is illustrated in figure 1.

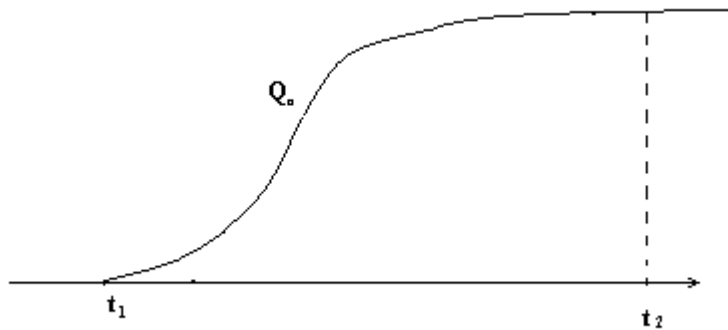


Figure 1. Probability distribution function

If check-in process begins at time t_c , and the bag-checking speed of EDS is a constant:

$$v(t) = \begin{cases} v & t_c \leq t < t_2 \\ 0 & \text{others} \end{cases} \quad (2)$$

Then the maximum number of checked bags

$$C(t) = \int_{-\infty}^t v(t)dt \quad (3)$$

Between t_c and t_2 $v(t)$ is a constant other is 0, so $C(t)$ is a segment showed in figure 2. Set time point after which there would be no queue up to checking ending time t_2 (Suppose that there are enough EDS so we have $t_p \leq t_2$). If passengers queue all the time between t_1 and t_p (which is illustrated in figure 3), that is the length of queue L has no zero point between t_1 and t_p .

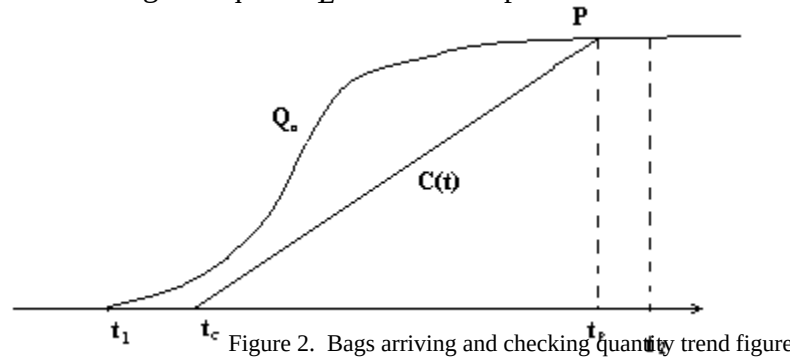


Figure 2. Bags arriving and checking quantity trend figure

Then the length of queue L could be computed using following function:

$$L(t) = QF(t) - C(t) \quad (4)$$

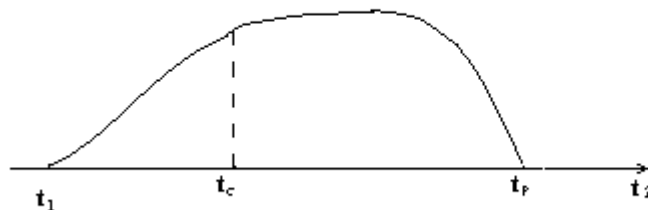


Figure 3. Queue length changing trend figure

Set $\bar{T}_w$ to be the average wait time for every passenger:

$$\bar{W} = \frac{1}{Q} \int_{t_1}^{t_2} L(t) dt \quad (5)$$

Set $T_{w\max}$ to be the maximum wait time for some passenger:

$$W_{\max} = \frac{\max(L(t))}{v} \quad (6)$$

Another Case

A special case is illustrated in figure 4 below that queue length $L(t)$ has zero points between t_1 and t_2 . In this case, the length of queue $L(t)$ is a complicated discontinues function that is showed in figure 4.

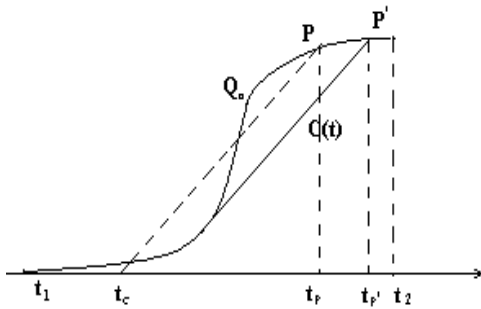
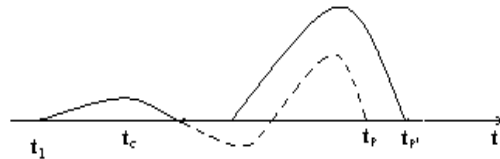


Figure 4. Number of arriving

increasing figure for special case.

Figure 5. Queue length function $L(t)$ in bags

for special case.

We find that under this case, few passengers arrive at the airport early, which delays t_p is to t_p and would cause them late for their flight. Actually, we could avoid this situation by suggesting passengers arrive earlier by manual or other ways. So we will ignore this case in our models.

Determination of Flight Schedule and Number of EDS

Interference between passenger flows

In order to develop our scheduling method, a model considering the interference of people in same queue while waiting for different flights is needed first. According our method, there is no interference among queues using different checking desks. But there might be interference among people that use the same checking desk.

Most passengers arrive at the airport within a time interval, and the arriving time of bags T is a random variable. Suppose that arriving time satisfy normal distribution. We consider two passenger flows' interference. Carefully set coordinate system to make $T_1 \sim N(-d, \sigma^2)$ and $T_2 \sim N(0, \sigma^2)$ with their probability density function are $f_1(t)$ and $f_2(t)$.

Figure 6. Interference of two different queues

In order to analyze interference of queues, we introduce interference degree I below:

$$I_{21}(d) = \int_{-\infty}^{\frac{d}{2}} f_2(t) dt \quad (7)$$

$$I_{12}(d) = 0$$

Later passenger flow would interfere an earlier one, but an earlier flow has no interference to later one. Interference $I_{21}(d)$ is controlled by time interval of two departure times. Derivate $I_{21}(d)$ with respect of d , we have:

$$I'_{21}(d) = -\frac{1}{2} f_2\left(\frac{d}{2}\right) < 0 \quad (8)$$

Since $I'_{21}(d) < 0$, we can conclude that interference degree $I'_{21}(d)$ decreases when the time interval d increases. When $d = 1$ hour, passengers typically arrive for their flight between departure times, thus we estimate $\sigma = 0.2803$, which could make sure 99.7% passengers arrive during forty-five minutes and two hours prior to their scheduled in probability. Thus the interference could be calculated:

$$I_{21}(1) = \int_{-\infty}^{\frac{1}{2}} f_2(t) dt \quad (9)$$

$$\begin{aligned} &= \int_{-\infty}^{\frac{1}{2}} \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{t^2}{2\sigma^2}\right] dt \\ &= 0.0082 \end{aligned}$$

There might be a queue 113 bags long, so the amount of passengers who are interfered might be $112.5 \times I_{21}(1) = 112.5 \times 0.0082 \approx 0.92$. Few passengers are interfered, thus we could ignore the interference when time distance is 1 hour.

● Number of EDSs Required by A Single Flight

The screening speed is a constant v . Set the number of bags needed screening is Q , screen time length is t , the number of EDSs needed:

$$n_{EDS} = \frac{Q}{vt} \quad (10)$$

● Grouping The Flights

Our goal is to minimize the expenditure of airports and the risk of explosions as well as average time on waiting. In fact, many factors such as density of passenger flow and the number of runways could affect the ways to schedule the departure of different types of flights within the peak hour.

Different airports use different methods to assign service counters to flights. A typical method is to assign one or more counters to one individual flight. In the case, passengers for same flight queue at the same counters assigned. Another commonly used method is to divide the service counters into several groups and each group is assigned to several different flights. The two methods both have advantages and disadvantages. If the EDS of a service counter does not work, it is easy for the group method to transfer passengers to other counters of the same group. However, because passengers for different flight wait in the same queue, there will be a disorder of priority. For example, the flight which Passenger A will take is going to depart in 30 minutes, but several passengers prior to him is going to take flight departing in 90 minute, later than that of A.

To utilize the advantages and meanwhile avoid the disadvantages, we also use the group method but each group is assigned to flights with the same departure time. Therefore, the advantages remain while the disorder of priority is negligible.

From assumption 2 and 3, we need to divide the flights in each airport into 12 groups and each group has at most 4 flights. Then we have the minimum number of EDS needed:

$$\min f = \sum_{j=1}^{12} \text{ceiling} \left(\sum_{i=1}^8 \frac{C_i r_i S_{i,j}}{v} \right) \quad (11)$$

and the restricts:

$$S_{i,j} = 0, 1, L \quad (12)$$

$$\sum_{j=1}^{12} S_{i,j} = M_i \quad (13)$$

$$\sum_{i=1}^8 S_{i,j} \leq 4 \quad (14)$$

where $S_{i,j}$ denotes the number of flights of type i and scheduled to take off at T_j and $\text{ceiling}(x)$ rounds the elements of x to the nearest integers towards infinity. We wrote a MATLAB program to solve this integer program problem, adapting both conservative and economical methods to both airports.

See appendix I to review our objectives and constrains in modeling.

● **Conservative schedule method**

To minimize the flight delay due to luggage screening, suppose that all flights fly with 100% of seats occupied, and each EDS examine 160 bags per hour. Thus

$$p_i=100\%, i=1,2,L,8$$

(15)

$$v=\frac{60}{160}=0.375 \text{ minutes per bag}$$

(16)

	:00	:05	:10	:15	:20	:25	:30	:35	:40	:45	:50	:55
M ₁	1	1		1	1	1		1	1	1	1	1
(34)												
M ₂	1	1					2					
(46)												
M ₃				1	1	1						
(85)												
M ₄			1								1	1
(128)												
M ₅		1	1	1	1	1	1	3	3	3	2	2
(142)												
M ₆			1	1	1	1	1					
(194)												
M ₇		1										
(215)												
M ₈	1											
(350)												
Total	430	437	464	455	455	455	428	460	460	460	446	446
EDSs	4	4	4	4	4	4	4	4	4	4	4	4

Table 1. Schedule of conservative method for airport A

	:00	:05	:10	:15	:20	:25	:30	:35	:40	:45	:50	:55
	2	1		1	1	1	1	1				

M ₁												
(34)												
M ₂			1	1					1	1	1	1
(46)												
M ₃		1	1		1				1	1	1	1
(85)												
M ₄			1			1	1	2				
(128)												
M ₅	1	1		1	1	1			1	1	1	1
(142)												
M ₆		1	1	1	1		2		1	1	1	1
(194)												
M ₇						1		1				
(215)												
M ₈	1											
(350)												
Total	560	455	453	416	455	519	550	505	467	467	467	467
EDSs	5	4	4	4	4	5	5	5	4	4	4	4

Table 2. Schedule of conservative method for airport B

● **Economical method**

Probably the flights would not fly with 100% of their seats occupied. Suppose that the flights fly with mean seat occupations and EDSs have mean rate of examination. Thus

$$p_i = \begin{cases} \frac{100\% + 70\%}{2} = 85\% & i = 1, 2, 3 \\ \frac{100\% + 60\%}{2} = 80\% & i = 4, 5, 6, 7 \\ \frac{100\% + 50\%}{2} = 75\% & i = 8 \end{cases} \tag{17}$$

$$v = \frac{\frac{60}{160 + 210}}{2} = 0.3243 \text{ minutes per bag} \tag{18}$$

	:00	:05	:10	:15	:20	:25	:30	:35	:40	:45	:50	:55
M ₁		1	1	1	1		1	1	1	1	1	1
(34)												
M ₂		1				2	1					
(46)												
M ₃			1	1	1							
(85)												
M ₄	1										1	1
(128)												
M ₅		1	1	1	1	1	1	3	3	3	2	2
(142)												
M ₆			1	1	1	1	1					
(194)												
M ₇	1											
(215)												
M ₈	1											
(350)												
Total	364	353	370	370	370	347	337	369	369	369	358	358
EDSs	3	3	3	3	3	3	3	3	3	3	3	3

Table 3. Schedule of economical method for airport A

	:00	:05	:10	:15	:20	:25	:30	:35	:40	:45	:50	:55
M ₁	2		1	1	1			1	1	1		
(34)												
M ₂		2				1	1				1	1
(46)												
M ₃	1		1	1	1	1	1					1
(85)												
M ₄								1	1	1	2	
(128)												

M ₅	1	1	1	1	1	1		1	1		1	
(142)												
M ₆			1	1	1	1	2	1	1	1	1	
(194)												
M ₇	1	1										
(215)												
M ₈	1											
(350)												
Total	393	364	387	370	370	380	380	442	400	400	399	380
EDSs	3	3	3	3	3	3	3	4	3	3	3	3

Table 4. Schedule of economical method for airport B

● Syntheses method

The conservative method costs too much and the economical method makes passengers wait for too long. Synthetically considering both advantages and disadvantages of each method above, we developed new schedule based on the conservative method. Some EDSs being reduced, there is a compromise between airport expenditure and efficiency.

	:00	:05	:10	:15	:20	:25	:30	:35	:40	:45	:50	:55
M_1	1	1		1	1	1		1	1	1	1	1
(34)												
M_2	1	1					2					
(46)												
M_3				1	1	1						
(85)												
M_4			1								1	1
(128)												
M_5		1	1	1	1	1	1	3	3	3	2	2
(142)												
M_6			1	1	1	1	1					
(194)												

M ₇	1											
(215)												
M ₈	1											
(350)												
Total	430	437	464	455	455	455	428	460	460	460	446	446
EDSs	3	3	4	3	4	3	3	4	4	4	3	3

Table 5. Schedule of syntheses method for airport A

	:00	:05	:10	:15	:20	:25	:30	:35	:40	:45	:50	:55
M ₁	2	1		1	1	1	1	1				
(34)												
M ₂			1	1					1	1	1	1
(46)												
M ₃		1	1		1				1	1	1	1
(85)												
M ₄			1			1	1	2				
(128)												
M ₅	1	1		1	1	1			1	1	1	1
(142)												
M ₆		1	1	1	1		2		1	1	1	1
(194)												
M ₇						1		1				
(215)												
M ₈	1											
(350)												
Total	560	455	453	416	455	519	550	505	467	467	467	467
EDSs	5	4	4	3	4	4	4	4	3	3	3	3

Table 6. Schedule of syntheses method for airport B

A memo on number of EDS determination and flight scheduling is available. See appendix II.

Evaluation of Model I

● Theoretical Method

Suppose bag arrival time T satisfies normal distribution, and then the distribution parameters could be estimated. Passengers arrive at the airport between two hours and 45 minutes prior their schedule, suppose most passengers arrive at the airport between those time intervals. Set two hour prior their schedule to be the origin of coordinate system, then we have $\mu = 0.625$ hour and $\sigma = 0.2083$. The probability density function $f(t)$:

$$f(t) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(t-\mu)^2}{2\sigma^2}\right) \quad (19)$$

The probability distribution function $F(t)$

$$F(t) = \int_{-\infty}^t f(t)dt \quad (20)$$

Then the length of queue L could be computed using following function:

$$L(t) = QF(t) - C(t) \quad (21)$$

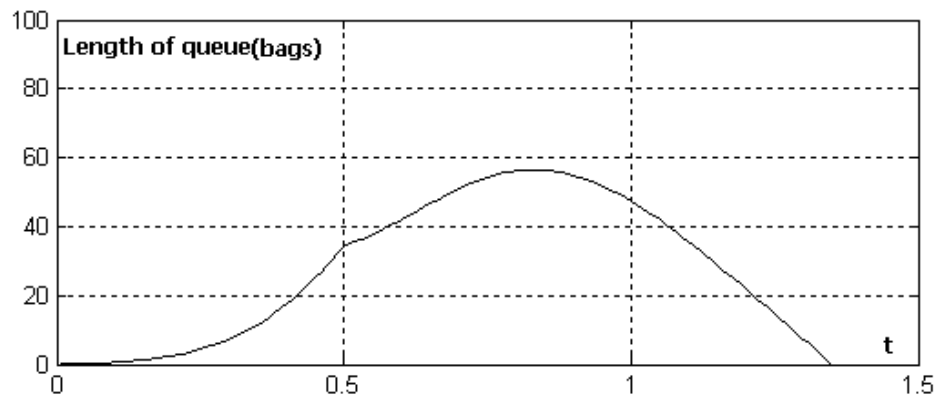


Figure 7. Plot of length of queue

Average waiting time is

$$\bar{W} = \frac{1}{Q} \int_0^{1.5} L(t)dt \quad (22)$$

Maximum waiting time is

$$W_{\max} = \frac{\max(L(t))}{v} \quad (23)$$

To get a more comfort expression we ignore the fact that $C(t)$ is discontinues but we consider it as continues temporarily. Derivate $L(t)$ with respect of t , and set

$$\frac{dL(t)}{dt} = 0 \quad (24)$$

That is

$$\frac{d[QF(t) - C(t)]}{dt} = 0 \quad (25)$$

i.e.

$$Qf(t) - nv = 0 \quad (26)$$

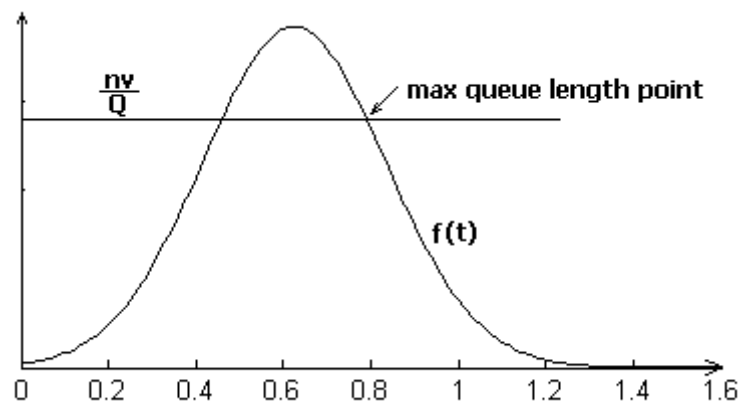


Figure 8. plot showing roots of equation (26)

Actually, we expect to deduce the cost, and then we must set $nv < \max[Qf(t)]$, then there must be two roots, but the bigger root $t_{\max}$ is which we are interested in. it is easy to confer that $\mu < t_{\max} < t_p$.

Then we have:

$$W_{\max} = \frac{L(t_{\max})}{v} \quad (27)$$

Departure Time	Amount of EDS	Average Longest Queue Time (minutes)	Average Length of Longest Queue (bags)	Average Maximum Wait (minutes)	Average Wait (minutes)
A	48	49.8	70.5	22.9	17.9
B	52	49.7	69.6	22.6	16.0

Table 7. Evaluation data obtained in theory

We write MATLAB programs to calculate the evaluation data using numerical method (see table 7). In order to see the worst status, we suppose that rates of seats occupied are 100% for all flight.

● Computer Simulation

Now that the flight schedule, the distribution of passenger flow and the regulation of check-in process are determined, we can simulate the check-in process using computer program. In computer simulation, we concentrate on average and maximum time spent on waiting for checking in. We wrote a MATLAB program to simulate both conservative and economical schedule strategies adapting to both airports. We simulate each case for 1,000 times to make the results statistically significant (algorithms and programs see appendix). The results of the simulation are shown below:

	EDSs needed	Average time on waiting (min/passenger)	Maximum time on waiting (min/passenger)	Average delayed time (min/flight)
Airport A, con	48	11.3402	15.9151	0
Airport B, con	52	11.1659	15.7682	0
Airport A, eco	36	18.4136	27.3442	0.5584
Airport B, eco	37	19.6329	29.5214	1.4614
Airport A, syn	41	14.8653	21.5671	0.1245
Airport B, syn	44	14.4732	20.8963	0.1098

Table 8. Computer simulation results for our schedule
(con: conservative; eco: economical; syn: syntheses)

See appendix IV to check the algorithms and programs of computer simulation.

Recommendation about checked baggage screening

During the peak hour, at least 46 flights depart at each airport, carrying up to 5,800 passengers. To follow the mandate of 100% screening of all checked bags while minimizing expenditure on purchasing and installing EDSs, we have several strategies.

● Conservative Strategy

If funds enough, we recommend to adopt the conservative strategy that to buy more EDSs so as to reduce passengers' time on waiting as much as possible. Theoretical estimation and computer simulation show that the average time spent on waiting for checking in is about 11 minutes per passenger and maximum time on waiting is lower than 16 minute. Furthermore, no delay caused by check in process will happen. These data show that airports adopting conservative strategy are quite efficient. Passengers would be satisfied with their efficiency.

● Economical Strategy

Each airport needs at least 48 EDSs to adopting conservative strategy, which would costs a lot of money. Moreover, during non-peak hour, some of the EDSs will be idle. If the primary goal is to save

money, we recommend the economical strategy that to purchase only 36 or 37 EDSs for each airport. Theoretical estimation and computer simulation show that the average time on waiting is between 18 and 20 minute and maximum time on waiting may be near 30 minute. Averagely, each flight will delay 0.5 to 1.5 minutes due to the check-in process. The time spent on queuing for passengers is a bit long, some passengers therefore may complain about waiting. The quality of service is not as good as it should be.

● Synthesis Strategy

Considering the advantages and disadvantages of each strategy, we should take the middle course. A Maximum Average Time (MAT) on queuing should be determined. Since Airports A and B and two largest facilities in the Midwest Region, the MATs should not exceed 15 minutes. So, there should be 41 EDSs for Airport A and 44 for Airport B. In this case, the average time on queuing is about 14.5 minutes and fewer than 10% flights could not finish check-in process 30 minute before departure. This should be acceptable by both passengers and airlines.

Model II: Enhanced Screening

Examination with both EDS and ETD

With the incorporation of ETD, the screening process needs modifying. Suppose that when an EDS is not operational to examine one bag, the bags will hand to an ETD. If an EDS determines one bag to be or possibly to be explosive, then this bag will also turn to an ETD. If an ETD is not operational to examine one bag or determine it to be explosive, this bag will be sent to security specialists for further examination. Thus all bags should be examined by EDS, and up to 20% bags should be complementarily examined by ETD. The 20% bags which need to be examined by ETD repeat a queue process with a new examine speed of 40 to 50 bags per hour per ETD.

● One ETD Checking Desk Service

As ETD screening is the second step for examining, there would be no queue at the ETD checking desk before starting screening. Meanwhile, if there is a queue at the EDS checking desk, then the bag arriving speed for ETD is a constant, so the probability density function is:

$$f(t) = \begin{cases} v_{\text{arriving}} & t_c < t < t_p \\ 0 & \text{others} \end{cases} \quad (28)$$

So the distribution of arriving bags for ETD system is mean. According to our basic queue model

If bag arriving time T satisfy mean distribution, that is all the bags arrive at the airport equably. Passengers typically arrive for their flight between forty-five minutes and two hours prior to their scheduled departure time, and then we could obtain its probability density function $f_m(t)$:

$$f(t) = \begin{cases} \frac{1}{t_2 - t_1} & t_1 < t < t_2 \\ 0 & \text{other} \end{cases} \quad (29)$$

And its probability distribution function $F_m(t)$:

$$F(t) = \begin{cases} 0 & t < t_1 \\ \frac{t-t_1}{t_2-t_1} & t_1 \leq t < t_2 \\ 1 & t \geq t_2 \end{cases} \quad (30)$$

Then we derive:

$$Q_a(t) = QF(t) = \begin{cases} 0 & t < t_1 \\ \frac{Q(t-t_1)}{t_2-t_1} & t_1 \leq t < t_2 \\ 0 & t \geq t_2 \end{cases} \quad (31)$$

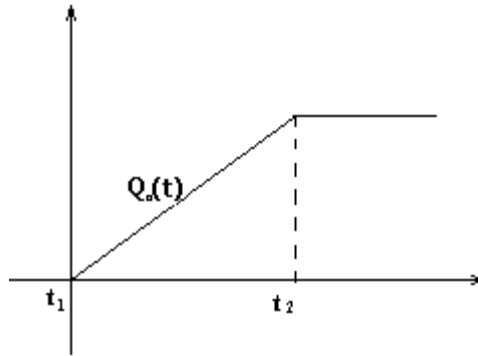


Figure 9. Number increasing trend
for mean distribution

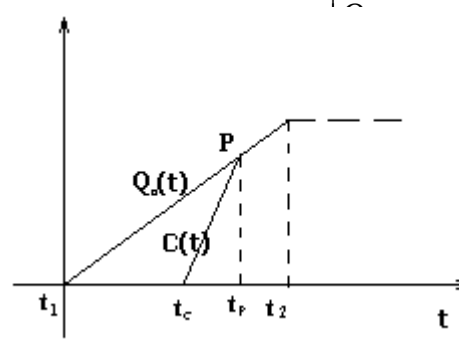


Figure 10. Number changing trend
when screening bags

Adapting the suppositions and the functions we have obtained, the length of queue L_m could be calculated out:

$$L(t) = \begin{cases} 0 & t < t_1 \\ \frac{Q(t-t_1)}{t_2-t_1} & t_1 \leq t < t_c \\ \frac{Q(t-t_1)}{t_2-t_1} - C(t) & t_c \leq t < t_p \\ 0 & t \geq t_p \end{cases} \quad (32)$$

The plot of function L is showed below:

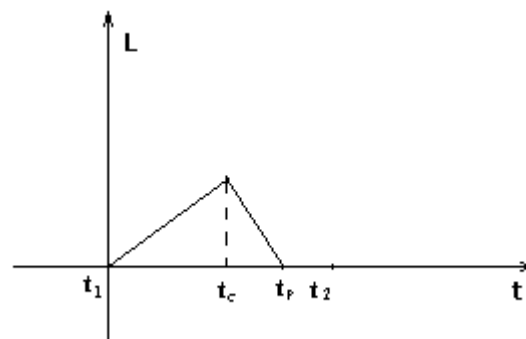


Figure 11. plot for queue length of mean distribution

Average wait is

$$\bar{W} = \frac{1}{Q} \int_{t_1}^{t_2} L(t) dt \quad (33)$$

Maximum wait is

$$W_{\max} = \frac{\max(L(t))}{v} = \frac{Q(t_c)}{v} \quad (34)$$

Adapt the mean distribution model to this case, and then we have:

$$L_{\max} = L(t_c) = 0 \quad (35)$$

According to discussions above, we can conclude that there would be no queue at the ETD checking desks, that is:

$$L(t) \equiv 0 \quad (36)$$

To insure all bags screening at time, we must set the screening speed v to satisfy:

$$v \geq v_{\text{arriving}} \quad (37)$$

● Could it be modeled as One ETD Checking Desk Service?

According to conclusions above, the interference would not come about. Then all ETD systems could be used as one checking desk. Moreover, passenger-wait at ETD checking desk is just the screening time about $\frac{1}{45}$ hour.

Set the quantity of working EDS is n_{EDS} , the screening speed of EDS is v_{EDS} . As about 20% bags need enhanced screening, so $v_{\text{arriving}} = 0.2v_{EDS}$. Set the screening speed of EDS is v_{EDS} , then the minimum quantity of ETD systems needed

$$n_{ETD} = \text{ceiling}\left(\frac{0.2n_{EDS}v_{EDS}}{v_{EDT}}\right) \quad (38)$$

Let $v_{EDS} = 185$ and $v_{ETD} = 45$ averagely, then

$$n_{ETD} = \text{ceiling}(0.82n_{EDS}) \quad (39)$$

Airport	Conservative	Economical	Syntheses
A	40	30	34
B	42	30	37

Table 9. Quantity of ETD needed for airport A and B

If we adopt conservative strategy, the average wait for all passengers is:

$$\bar{W}_A = 0.80 \times 11.34 + 0.20 \times (11.34 + 60 / 45) = 11.62 \text{ minutes}$$

$$W_{Amax} = 0.8 \times 15.91 + 0.2 \times (15.91 + 60 / 45) = 16.18 \text{ minutes}$$

$$\bar{W}_B = 0.80 \times 11.17 + 0.20 \times (11.17 + 60 / 45) = 11.43 \text{ minutes}$$

$$W_{Bmax} = 0.8 \times 15.91 + 0.2 \times (15.91 + 60 / 45) = 16.17 \text{ minutes}$$

See appendix III to review the memo of enhanced screening policy

Weaknesses of the Models

- Since we use greedy algorithm for grouping the flight, the schedule may not be most optimum.
- Theoretical evaluation is based on bag flows while computer simulations concentrate on passenger flow. The results will be slightly different.

Reference

[1] www.azworldairports.com

List of Appendix

Appendix I	Objectives and Constrains
Appendix II	Number of EDS Determination and Flight Scheduling
Appendix III	Memo of Enhanced Screening Policy
Appendix IV	Algorithms and Programs of Computer Simulation

Appendix I

Objectives and Constrains

Considering the safety of flights, we constrain that the peak hour is divided into 12 departure intervals with 5 minutes long. And the number of flights departing during 5 minutes should not exceed 4.

To make the 100%-screening system work more effective and economical, we develop several

different schedules to consult. In these schedules, all flights are considered to be theoretically full for investigating the worst situation. It relates to some factors such as queue length, average passengers' waiting time and maximum passengers' waiting time. Furthermore, we derive the time when the queue is longest in the model.

Among these factors, device accuracy p is an important factor. It would affect the amount of bags which need rechecking under a risk loss. According to some references in pattern recognizing, the amount of bags needed rechecking could be controlled by device accuracy, risk-probability and some other data. The higher accuracy, the fewer bags that need to be rechecked. Another similar parameter is error probability p_e which equals to $1 - p$. When the accuracy is high enough to up to 100%, it is more comfortable to evaluate the system performance by error probability p_e . As p is nearly up to 100%, accuracy is hard to promote a little but error probability could reduce several scalar grades.

Checking cost is another important factor. Whether new device could be widely applied is determined by its cost. The cost must be low enough to afford. Taking ETD for example, this kind of device is costly, so they are used restrictively to check some doubtful bags. A cheaper type EDS is used widely now in the airports.

Checking speed is also important. If checking speed is fast, we could use fewer devices and shorten the length of queue.

Appendix II

Number of EDS Determination and Flight Scheduling

The number of EDS used by an airport depends on the density of passenger flow during the peak hour and the examination speed of EDS. The flight schedule can also affect the number of EDS.

To determine the number of EDS and the flight schedule, follow the steps below:

1. Investigate data of flights departing during peak hour. Flight types, number of seats on each type of flight, number of flights of each type, average occupation of seats are required.
2. Estimate the number of EDS needed for each flight.
3. According to the interval of departing (e.g. 5 minutes), divide all the flights into several groups (e.g. 12). Compute the sum of numbers of EDSs needed for each flight of same group and round the sum to the nearest integer towards infinity. That is the number of EDS needed for this group.
4. Modify the grouping to make the total amount of EDSs minimum.
5. Assign departure time for each group of flight. Flights in same group depart at the same time.
6. Allocate service counters (each counter has one EDS) for each group. Flights in same group use same counters.

Appendix III

Memo of Enhanced Screening Policy

We give a technical analysis to the enhanced screening policy in a method detailed below. The analysis shows that the cost of using ETD together with EDS is justified in light of the value that it provides. Economically speaking, EDS devices shouldn't be replaced by ETD. And according to our analysis, it is expensive to use ETD separately. The cost is 18 times than that of EDS, but the error probability of ETD used separately is 1/5 times that of EDS. However, if we use ETD and EDS together, the cost is 19 times than that of using EDS separately, the error probability is 3/1000 of using EDS separately.

1. Using ETD Separately

To evaluate performance of EDS and ETD when they are used separately, we consider the cost for checking one bag and error probability.

Given that each EDS costs \$1 million and \$100 thousand to install in an airport, and each EDS could be used 16 hours each day for 10 years averagely. Then the cost of hard ware for each EDS would

be $\frac{1 \times 10^6 + 1 \times 10^5}{10 \times 365 \times 16} \approx 18.8356$ dollars/hour. The labor cost would be \$15/hour if each EDS just need one

worker to management. It takes $\frac{1}{185} \approx 0.0054$ hour to check one bag if each EDS checks 185 bags per

hour. Then the total cost for EDS would be $m_{EDS} = (18.85 + 15) \times 0.0054 = 0.1828$ dollars per bag. In the

same way we derive $m_{EDT} = (2.1404 + 15 \times 10) / 45 = 3.3809$ dollars per bag.

The error probability for EDS is $p_{eEDS} = 1.5\%$, while that of ETD $p_{eEDT} = 0.3\%$.

Comparing the cost for checking one bag and the error probability, the cost of ETD is

$\frac{3.3809}{0.1828} \approx 18.1012$ times than that of ETD while the error probability of ETD is $\frac{1.5\%}{0.3\%} = 5$ than times that of EDS.

According data we have derived, it is not economical to use ETD separately.

2. The Cooperation of EDSs and ETDs

Using the same type of device couldn't decrease error probability. In order to decrease it, we need the two types of devices to work together.

If we use EDS only, the cost for checking one bag would be 0.1828 dollars, and the error

probability would be 1.5%.

If we use ETD together with EDS, the cost for checking one bag would be $0.1828 + 3.3809 = 3.5637$ dollars per bag, which is 19.1012 times that of using EDS only. The error probability would be $p_e = p_{eEDS} \times p_{eEDT} = 4.5 \times 10^{-5}$ while the error probability for using EDS only are 333 times that of using the two types together.

From data we obtained above, we could conclude that the cost of using EDS and ETD together is justified in light of the value that it provides.

Recommendation on Research Funding

To best fund future scientific programs, we consider several aspects, including device technology, cost, accuracy, speed, operational reliability and research in STEM.

Generally speaking, we should promote device technology, accuracy, speed and operational reliability, and reduce cost. Among factors we have mentioned, emphasis should be placed on technology, cost and speed. Accuracy and operational reliability are not so important.

Device Technology

As error probability couldn't be simply deduced by check bags several times using the same kind of device, more new type of check-device is needed. So, we should expend more money on device technology. Taking example for ETD device, the error probability could decrease about 100 times when using ETD together with EDS.

Cost and Speed

Cost and speed have the biggest effect on the application of device. According to our analysis, the reason why ETD system couldn't replace EDS is that using ETD system separately is too costly. Because of ETD's high labor cost and low checking speed, the check-one-bag cost is 18 times that of EDS. So, widely using of ETD system is constrained.

Accuracy and Operational Reliability

Because device accuracy and operational reliability is nearly up to 100%, it is costly and hard to promote them. And even if we promote them a little, there would be a slight effect on security and check efficiency.

Research in STEM Areas

In STEM areas, PATTERN RECOGNITION and AUTOCONTROL have the biggest impact on

security system performance.

In security systems, some passengers would need to check their bags twice or more times. According to references of PATTERN RECOGNITION, rate of several-time-check is controlled by risk loss. If we have good determination method, the rate would be reduced.

As we mentioned above, widely using of security system is constrained by cost. Take ETD system for example, ETD system is costly. Labor cost is the main part of all costs for ETD system. So we need more deeply research in AUTOCONTROL area.

Appendix IV

Algorithms and Programs of Computer Simulation

Greedy Algorithms for Flight Grouping

Sort the flight by the number or seats, descendingly. To each flight, check all 12 departure times to find the time with lowest number of passengers. If the time is assign no more than 3 flights to depart, assign the flight to this departure time.

Simulation of All Flights During the Peak Hour

Produce 3 groups of random numbers to simulate occupations of each flight, passengers' arrival time, and number of bags taken by each passenger, respectively. Compute and memorize times spent on waiting of each passenger and delayed times for each flight during the peak hour. Based on the data simulated and computed above, evaluate the average time on waiting, maximum time on waiting and average time delayed. Repeat this procedure until the results are statistically significant.

```
function Ta=Taverage(Q,n,v)
%=====
% main function for solve average wait
%=====
Ta=intlengthh(1.5,Q,n,v)/Q;
return;

function L=lengthh(t,Q,n,v)
%=====
% calculate the length of a given queue at time t
%=====
L=0;
if t<=0|t>=1.5
    L==0;
elseif t>0&t<=0.5
    L=intfnorm(t,Q);
elseif t>0.5&t<1.5
    L=intfnorm(t,Q)-intV(t,n,v);
```

```

end
if L<0
    L=0;
end
return

```

```

function s=intlengthh(b,Q,N,v)
%=====
% integral funtion for queue length
%=====
eps=1.0e-2;
a=0;
t1=10000;
t2=-10000;
n=2;
t(1,1)=0.5*(b-a)*(lengthh(a,Q,N,v)+lengthh(b,Q,N,v));
while abs(t2-t1)>=eps
    area=0.0;
    %n=n+1
    h=(b-a)/2^(n-1);
    for i=1:(2^(n-1))
        area=area+0.5*h*(lengthh(a+h*(i-1),Q,N,v)+lengthh(h*i+a,Q,N,v));
    end
    t(n,1)=area;
    for j=2:n
        for i=1:(n-j+1)
            t(i,j)=(4^(j-1)*t(i+1,j-1)-t(i,j-1))/(4^(j-1)-1);
        end
    end
    t1=t(1,n);
    t2=t(1,n-1);
    n=n+1;
end
s=t1;
return;

```

```

function y=Maxmy(Q,n,v)
%=====
%function for slove the max length of queue(Lmax),
%what time the queue is longest(tmax) and max wait(Twaitmax)
%=====
miu=0.625;

```

```

thegema=miu/3;
tmax=miu+sqrt(-2*thegema^2*log(sqrt(2*pi)*n*v*thegema/Q));
Lmax=lengthh(tmax,Q,n,v)/n;
Twaitmax=Lmax/v;
y=[tmax,Lmax,Twaitmax];

```

```

function y=intV(t,n,v)
%=====
% calculate checked quantity at time t
%=====
y=0;
if t>=0.5&t<=1.5
    y=n.*v.*(t-0.5);
end
return;

```

```

function s=intfnorm(b,Q)
%=====
% integrate the function fnorm
%=====
eps=1.0e-3;
a=zeros(size(b));
t1=10000;
t2=-10000;
n=2;
t(1,1)=0.5.*(b-a).*(fnorm(a)+fnorm(b));
while abs(t2-t1)>=eps
    area=0.0;
    %n=n+1
    h=(b-a)/2^(n-1);
    for i=1:(2^(n-1))
        area=area+0.5.*h.*(fnorm(a+h.*(i-1))+fnorm(h.*i+a));
    end
    t(n,1)=area;
    for j=2:n
        for i=1:(n-j+1)
            t(i,j)=(4.^(j-1).*(t(i+1,j-1)-t(i,j-1))./(4.^(j-1)-1);
        end
    end
    t1=t(1,n);
    t2=t(1,n-1);

```

```

n=n+1;
end
s=t1*Q;
return;

```

```

function y=fnorm(x)
%=====
% Normal distribution probability density function
%=====
miu=0.625;
thegema=0.2038;
y=exp(-((x-miu)/thegema).^2/2)./sqrt(2.*pi)./thegema;

```

```

%flightarrangement.m
%=====
%group the flights using greedy method
%=====
schstat=zeros(12,5);
bigflinow=48;
shtqunow=1;
for i=48:-1:1
    shtqunow=findshtqu(schstat);
    schstat=addfli(schstat,shtqunow,i,Aflightcorr);
end

```

```

%=====
% find the group with smallest number of passengers
%=====
function y=findshtqu(sch)
m=10000;
for i=1:12
    if sch(i,5)<m & sch(i,4)==0
        m=sch(i,5);
        y=i;
    end
end
end

```

```

%=====
% add flight No.n to group shtqu
%=====

```

```

function y=addfli(schstat,shtqu,n,flighttype)
y=schstat;
for i=1:4
    if schstat(shtqu,i)==0
        y(shtqu,i)=n;
        y(shtqu,5)=y(shtqu,5)+flighttype(n);
        break
    end
end

%=====
% queue simulation
%=====

function [averwaittime,averdelaytime,probdelay,maxqulen,maxwaitpoint]=fs1(sch)
totwt=0; % data initiation
totdt=0;
totpd=0;
totmql=0;
totmwp=0;
for sims=1:1000 % loop 1000times
    waittime=0;
    delaytime=0;
    pd=0;
    mql=0;
    mwp=0;
    for i=1:12
        mqlnow=0;
        mwpnow=0;
        stat=prework(sch); % produce random seat occupation
        passnum=ceil(stat(i,5)); % compute number of passengers
        arrvtime=sort(randn(1,passnum)*.2083+.625)*60; %produce random passenger
        for k=1:passnum %arrival time
            if arrvtime(k)<30
                arrvtime(k)=30;
            end
        end
        wtlast=0;
        luggrand=rand(1,1); %produce random number of bags for one passenger
        num=stat(i,7);
        if luggrand<.2
            luggnum=0;
            wtlast=0;

```

```

elseif luggrand<.4
    luggnum=1;
    wtlast=60/num./(rand(1,luggnum)*50+160);
else
    luggnum=2;
    wtlast=sum(60/num./(rand(1,luggnum)*50+160)); %time spent on waiting
end                                     %of the first passenger
waittime=waittime+wtlast;
for n=2:passnum
    luggrand=rand(1,1);
    if luggrand<.2
        luggnum=0;
        wtthis=wtlast-(arrvtime(n)-arrvtime(n-1));
    elseif luggrand<.4
        luggnum=1;          wtthis=wtlast+60/num./(rand(1,luggnum)*50+160)-(arrvtime(n)-arrvtime(n-1));
    else                    %compute waiting time using recursion
        luggnum=2;          wtthis=wtlast+sum(60/num./(rand(1,luggnum)*50+160))-(arrvtime(n)-arrvtime(n-
1));    end                    %compute waiting time using recursion
    waittime=waittime+wtthis;
    wtlast=wtthis;
    if wtthis>mqlnow
        mqlnow=wtthis;
        mwpnow=arrvtime(n);
    end
end
if arrvtime(passnum)+wtlast<=90
    dlthis=0;
else
    dlthis=arrvtime(passnum)+wtlast-90;
    pd=pd+1;
end
delaytime=delaytime+dlthis;
mql=mql+mqlnow;
mwp=mwp+mwpnow;
end
mql=mql/12; average
mwp=mwp/12;
delaytime=delaytime/12;
pd=pd/12;
waittime=waittime/sum(stat(1:12,5));
totmql=totmql+mql;    add
totmwp=totmwp+mwp;
totwt=totwt+waittime;

```

```
    totdt=totdt+delaytime;
    totpd=totpd+pd;
end
averwaittime=totwt/1000; average
averdelaytime=totdt/1000;
probdelay=totpd/1000;
maxqulen=totmql/1000;
maxwaitpoint=totmwp/1000;

%=====
% produce random seat occupation
%=====
function y=prework(stat)
y=stat;
for i=1:12
    for j=1:4
        if y(i,j)<=85
            y(i,j)=y(i,j)*(rand(1)*.3+.7); uniform distribution
        elseif y(i,j)<=215
            y(i,j)=y(i,j)*(rand(1)*.4+.6);
        else
            y(i,j)=y(i,j)*(rand(1)*.5+.5);
        end
    end
    y(i,5)=sum(y(i,1:4));
end
```